ExamGOAL
PYQ PDF

Free · With answer key

JEE MAIN · MATHEMATICS · CHAPTER-WISE PYQS

Logarithm

JEE Main Mathematics · Previous Year Questions with Answer Key

11	11	2020 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

Numerical	6 questions
MCQ (Single Correct Answer)	5 questions
Answer key	at the end

Practise online

Every question with its full solution, free on **questions.examside.com**

1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.
3. **Understand**
   Scan the QR code beside any question for its full solution.

Numerical 6 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

The sum of squares of all the real solutions of the equation

$\log_{(x+1)}\left(2x^2+5x+3\right)=4-\log_{(2x+3)}\left(x^2+2x+1\right)$ is equal to ____ .

Open solution

Q2. JEE Main 2023 (Online) 10th April Morning Shift

Let a, b, c be three distinct positive real numbers such that $(2a)^{\log_e a}=(bc)^{\log_e b}$ and $b^{\log_e 2}=a^{\log_e c}$.

Then, 6a + 5bc is equal to _____.

Open solution

Q3. JEE Main 2023 (Online) 25th January Morning Shift

Let $S=\left\{\alpha:\log_2(9^{2\alpha-4}+13)-\log_2\left(\frac{5}{2}.\ 3^{2\alpha-4}+1\right)=2\right\}$. Then the maximum value of β for which the equation $x^2-2\left(\sum_{\alpha\in S}\alpha\right)^2 x+\sum_{\alpha\in S}(\alpha+1)^2\beta=0$ has real roots, is _____.

Open solution

Q4. JEE Main 2021 (Online) 20th July Evening Shift

The number of solutions of the equation

$\log_{(x+1)}(2x^2+7x+5)+\log_{(2x+5)}(x+1)^2-4=0$, x > 0, is :

Open solution

Q5. JEE Main 2021 (Online) 26th February Morning Shift

The number of solutions of the equation $\log_4(x-1)=\log_2(x-3)$ is _____.

Open solution

Q6. JEE Main 2020 (Online) 9th January Morning Slot

The number of distinct solutions of the equation
$\log_{\frac{1}{2}}|\sin x|=2-\log_{\frac{1}{2}}|\cos x|$ in the interval [0, 2π], is ____.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

MCQ (Single Correct Answer) 5 questions

Q7. JEE Main 2026 (Online) 4th April Evening Shift

Let $\alpha = \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \ldots \infty$ and

$\beta = \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \ldots \infty$. Then the value of

$(0.2)^{\log_{\sqrt{5}}(\alpha)} + (0.04)^{\log_5(\beta)}$ is equal to :

(A) 4 (B) 5 (C) 8 (D) 25

Open solution

Q8. JEE Main 2026 (Online) 23rd January Evening Shift

The sum of all the real solutions of the equation
$\log_{(x+3)}\left(6x^2 + 28x + 30\right) = 5 - 2\log_{(6x+10)}\left(x^2 + 6x + 9\right)$ is equal to :

(A) 1 (B) 4 (C) 0 (D) 2

Open solution

Q9. JEE Main 2025 (Online) 22nd January Morning Shift

The product of all solutions of the equation $e^{5(\log_e x)^2+3} = x^8, x > 0$, is :

(A) e^2 (B) e (C) $e^{6/5}$ (D) $e^{8/5}$

Open solution

Q10. JEE Main 2023 (Online) 11th April Morning Shift

The number of integral solutions x of $\log_{(x+\frac{7}{2})}\left(\frac{x-7}{2x-3}\right)^2 \geq 0$ is :

(A) 8 (B) 7 (C) 5 (D) 6

Open solution

Q11. JEE Main 2023 (Online) 30th January Morning Shift

If the solution of the equation $\log_{\cos x}\cot x + 4\log_{\sin x}\tan x = 1, x \in \left(0, \frac{\pi}{2}\right)$, is $\sin^{-1}\left(\frac{\alpha+\sqrt{\beta}}{2}\right)$, where α, β are integers, then $\alpha + \beta$ is equal to :

(A) 3 (B) 6 (C) 4 (D) 5

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Answer Key

Numerical

1. **2**	2. **8**	3. **25**	4. **1**	5. **1**
6. **8**				

MCQ (Single Correct Answer)

7. **C**	8. **C**	9. **D**	10. **D**	11. **C**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

ExamGOAL
PYQ PDF

Your Exam. Our Responsibility.

From first chapter to exam day.

Practise with the questions your exam has already asked, then test yourself under real exam conditions.

PAST YEAR QUESTIONS

Every past paper, chapter-wise.

Previous year questions sorted the way you study, each with a full solution.

- Chapter-wise and year-wise
- Weightage and topic tags on every chapter
- Your progress on each one, at a glance

MOCK TESTS

Mock tests with a real exam feel.

Practise under the same pressure as the real paper, on your phone or on the big screen.

- Timed sections, just like exam day
- Mark questions for review and come back

ANALYSIS

Know your score. Know your rank.

A full report after every test, so you know exactly where you stand.

- Your percentile among everyone who took it
- Predicted real-exam percentile and All India Rank
- Subject breakdown and time analysis

PRACTICE

Solve. Check. Understand.

Learn from every question, not just the ones you get right.

- Instant answers as you practise
- Step-by-step explanations
- See how everyone else answered

NCERT Exemplar
With full solutions

Exam syllabus
Subject and chapter-wise

Bookmarks
Organised in groups

Your notes
On any question

ExamGOAL
LEARNING APP

ExamGOAL: Exam Preparation App

One app for every exam. Free trial for new users, on Android and the web.

GET IT ON Google Play

Scan to get the app

www.examgoal.com
Practice and mock tests

questions.examside.com
Free solutions

pdf.questions.examgoal.com
Free PYQ PDFs